

Important Topics

- **Self exploration**
- **guidelines for value education**
- **utilitarian theory**
- **precondition desires and thoughts and problems**
- **value education for better society**
- **the challenger**
- **insider trading**
- **industrial standards**
- **ethical theories**
- **IPR and infringement**
- **religion and self respect**
- **collective bargaining**
- **expert witness**
- **environmental ethics**
- **computer ethics**

Types of Values

- Right Conduct

- (a) SELF-HELP SKILLS: Care of possessions, diet, hygiene, modesty, posture, self reliance, and tidy appearance
- (b) SOCIAL SKILLS: Good behavior, good manners, good relationships, helpfulness, No wastage, and good environment, and
- (c) ETHICAL SKILLS: Code of conduct, courage, dependability, duty, efficiency, ingenuity, initiative, perseverance, punctuality, resourcefulness, respect for all, and responsibility

- Peace

- Attention, calmness, concentration, contentment, dignity, discipline, equality, equanimity, faithfulness, focus, gratitude, happiness, harmony, humility, inner silence, optimism, patience, reflection, satisfaction, self-acceptance, self-confidence, self-control, self-discipline, self-esteem, self-respect, sense control, tolerance, and understanding

- Truth

- Accuracy, curiosity, discernment, fairness, fearlessness, honesty, integrity (unity of thought, word, and deed), intuition, justice, optimism, purity, quest for knowledge, reason, self-analysis, sincerity, sprit of enquiry, synthesis, trust, truthfulness, and determination.

- Love

- Acceptance, affection, care, compassion, consideration, dedication, devotion, empathy, forbearance, forgiveness, friendship, generosity, gentleness, humanness, interdependence, kindness, patience, patriotism, reverence, sacrifice, selflessness, service, sharing, sympathy, thoughtfulness, tolerance and trust

- Non – violence

- (a) PSYCHOLOGICAL: Benevolence, compassion, concern for others, consideration, forbearance, forgiveness, manners, happiness, loyalty, morality, and universal love
- (b) SOCIAL: Appreciation of other cultures and religions, brotherhood, care of environment, citizenship, equality, harmlessness, national awareness, perseverance, respect for property, and social justice.

Answer Type Questions

Name a few basic guidelines for value education?

The subject that enables us to understand 'what is valuable' for human happiness is called value education. In order to qualify for any course on value education, the following guidelines for the content of the course are important.

Given below few guidelines:

1. Universal
2. Rational
3. Natural and verifiable
4. All-encompassing
5. Leading to harmony

What are the basic guidelines for value education? (RTU 2017 - 18)

The subject that enables us to understand 'what is valuable' for human happiness is called value education. In order to qualify for any course on value education, the following guidelines for the content of the course are important:

Universal: It needs to be applicable to all human beings irrespective of caste, creed, nationalities, religion, etc., for all times and regions.

Rational: It has to appeal to human reasoning. It has to be amenable to reasoning and not based on dogmas or blind beliefs.

Natural and verifiable: It has to be naturally acceptable to the human being who goes through the course and when we live on the basis of such values it leads to our happiness. It needs to be experientially verifiable, and not based on dogmas, beliefs or assumptions.

and real human life and profession. The value education among human beings and with nature.

Leading to harmony: The value education among human beings and with nature, harmony within the individual, among human beings and with nature.

Long Question Answers

Q.1. What is the need for value education?

OR

Write a short note on the need for value education in today's (RTU 2016-17)

Ans.

Write a short note on the need for value education in today's (RTU 2016-17)

Correct identification of our aspirations: The subject which enables us to understand 'what is valuable' for human happiness is called 'value education' (VE). Thus, VE enables us to understand our needs and visualize our goals correctly and also indicate the direction for their fulfillment. It also helps to remove our confusions and contradictions and bring harmony at all levels.

Understanding universal human values to fulfil our aspirations in continuity: Values form the basis for all our thoughts, behaviours and actions. Once we know what is valuable to us, these values become the basis, the anchor for our actions. We also need to understand the universality of various human values, because only then we can have a definite and common program for value education. Then only we can be assured of a happy and harmonious human society.

Complementarity of values and skills: To fulfil our aspirations both values and skills are necessary. When we identify and set the right goals and produce in the right direction. This is known as the value domain, the domain of wisdom, and when we learn and practice to actualize this goal to develop the techniques to make this happen in real life, in various dimensions of human endeavour (struggle). This is known as the domain of skills.

Hence, there is an essential complementarity between values and skills for the success of any human endeavour. For example, I want to lead a healthy life. Only wishing for good health will not help me keep my body fit and healthy and without having understood the meaning of health, I will not be able to choose things correctly to keep my body fit and healthy.

Re-evaluation of our beliefs: Each one of us believes in certain things and we base our values on these beliefs, are they false or true which may or may not be true in reality. These beliefs come to us from what we read, see, hear, what our parents tell us, our friends talk about, what the magazines talk of, what we

see from TV etc. Value Education helps us to evaluate our beliefs and assumed values.

Technology and human values: The present education system has become largely skill-based. The prime emphasis is on science and technology. However, science and technology can only help to provide the means to achieve what is considered valuable. It is not within the scope of science and technology to provide the competence of deciding what really is valuable. Value Education is a crucial missing link in the present education system. Because of this deficiency, most of our efforts may prove to be counterproductive and serious crises at the individual, societal and environmental level are manifesting.

2. Define Self-exploration. What is the content of self - exploration?

Self-exploration is the process to find out what is valuable to me by investigating within myself, what is right for me, true for me, has to be judged within myself. Through self-exploration we get the value of ourselves. We live with different entirety (family, friends, air, soil, water, trees, etc.) and we want to understand our relationship with all these. For this we need to start observing inside. The main focus of self-exploration is myself - the human being. Content of self-exploration is just finding answers to the following fundamental questions of all human beings:

1. **The Desire/Goal:** What is my (human) Desire/ Goal? What do I really want in life, or what is the goal of human life?
2. **Program:** What is my (human) program for fulfilling the desire? How to fulfil it? What is the program to actualize the above?

In short, the above two questions cover the whole domain of human aspirations and human endeavour. Thus, they form the content of self-exploration.

Directions of happiness in the society

Short Answer Type Questions

Q.1. What is pre-conditioning? What is their source?

or

What is the meaning of desire? How do we verify whether our desires are coming from sensations or preconditioning or natural acceptance?

or

How does the human mind get influenced or conditioned?

Ans. Preconditioning means we have assumed something about our desires on the basis of prevailing notions about it. They come from what we read, see, hear, what our parents tell us, our friends talk about what the magazines talk of, what we see on the TV etc. We have not self-verified the desires in our own right. As a result, we are not clear about what we will get out of fulfillment of that desire. The problem with that is, unless we verify our desires, we may not even know whether they are ours. We may end up spending an entire lifetime accumulating desires that are not ours, and in running about trying to fulfill them.

Q.2. Suggest any two programs that you can undertake to improve the health of your body.

Ans. To ensure the health of the body, we need to undertake the following programs: (any two)

1. Proper upkeep of the body: When we work, the body gets tired. When we rest, the body becomes fit to work. So we need to ensure proper time, posture and ways to work and to rest. These issues are included in the upkeep of the body.

2. Physical labour and exercise: Requisite amount of physical labour and exercise are essential to keep the body healthy. Labour means employing the body physically for production and maintenance of physical facilities.

3. Asan-Pranayam: Yogasana and Pranayama are well-designed exercises to keep the body healthy and to ensure the synergy between self and the body. These exercises involve specific postures and regulations of breathing.

Q.3. Define Existence and Co-existence.

Ans. Co-existence in nature means there is a relationship and complementarity among all the entities in nature including human beings. Co-existence is a state in which two or more groups are living together while respecting their differences and resolving their conflicts non-violently. To exist together (in time or space) and to exist in mutual tolerance.

ENGINEERS AS MANAGERS

Characteristics

Characteristics of engineers as managers are:

1. Promote an ethical climate, through framing organization policies, responsibilities and by personal attitudes and obligations.
2. Resolving conflicts, by evolving priority, developing mutual understanding, generating various alternative solutions to problems.
3. Social responsibility to stakeholders, customers and employers. They act to develop wealth as well as the welfare of the society. Ethicists project the view that the manager's responsibility is only to increase the profit of the organization, and only the engineers have the responsibility to protect the safety, health, and welfare of the public. But managers have the ethical responsibility to produce safe and good products (or useful service), while showing respect for the human beings who include the employees, customers and the public. Hence, the objective for the managers and engineers is to produce valuable products that are also profitable.

Managing Conflicts

Managing conflicts, force should not be resorted. In fact, the conflict situations should be tolerated, avoided, and resolved by participation by all the concerned. The conflicts in case of project managers are the following manners:

1) Conflicts based on schedules: This happens because of various levels of execution, priority and limitations of each level.

2) Conflicts arising out of fixing the priority to different projects or departments. This is to be arrived at from the end requirements and it may change from time to time.

3) Conflict based on the availability of personnel.

4) Conflict over technical, economic, and time factors such as cost, time, and performance level.

5) Conflict arising in administration such as authority, responsibility, accountability, and logistics required.

Conflicts of personality, human psychology and ego problems.

Conflict over expenditure and its deviations.

Most of the conflicts can be resolved by following the principles listed here:

...all concerned should be obtained. The

circumstances.

4.6 SAFETY LESSONS FROM 'THE CHALLENGER'

The safety lessons one can learn in the Challenger case are as follows:

1. Negligence in design efforts. The booster rocket casing recovered from earlier flights had the failure of filed-joint seals. No design changes were incorporated. Instead of two or three rings should have been fixed. But there was no time for testing with three or at least three rings could have been tried while launching.
2. Tests on O-rings should have been conducted down to the expected ambient temperature, i.e., to 20 °F. No normalization of deviances should have been allowed.
3. NASA was not willing to wait for the weather to improve. The weather was not ideal on the day of launch. A strong wind shear might have caused the rupture of the weak O-rings.
4. The final decision making of launch or no-launch should have been with the engineers, not on the managers. Engineers insisted on 'safety' but the managers went ahead with the 'schedule'.
5. Informed consent: The mission was full of dangers. The astronauts should have been informed of the probable failure of the O-rings (field joints). No informed consent was obtained when the engineers had expressed that the specific launch was unsafe.
6. Conflict of interest (Risk Vs. Cost): There were 700 criticality-1 items, which included field joints. A failure in any one of them would have caused the tragedy. No backup plan had been provided for these criticality-1 components.
7. Escape mechanism or 'safe exit' should have been incorporated in the craft.

5.0 GLOBALIZATION

Globalization means integration of countries through commerce, transfer of technology, of information and culture. In a way, it includes acting together and interacting economically. Trade, investment, loan, development schemes and capital across countries. In a different way, flows include knowledge, science, technology, skills, culture, information, and entertainment. Direct human resource, tele-work, and outsourcing. This interdependence has increased tensions and ruptures among the nations. For the engineers, the issues such as multinational computer, internet functions, military development and environmental ethics have gained importance for their very sustenance and progress.

5.1 MULTINATIONAL CORPORATIONS

Organisations who have established business in more than one country, are called multinational corporation. The headquarters are in the home country and the business is extended in other countries. The Western organizations doing business in the less-economically developed (and overpopulated) countries gain the advantage of inexpensive labor, availability of natural resources, conducive-tax atmosphere, and virgin market for the products. At the same time, the developing countries are also benefited by fresh job opportunities, jobs with higher remuneration and challenges, technology, and several social benefits by the wealth developed. But this happens invariably at the cost of social and cultural disturbance. Loss of jobs for the home country, and loss or exploitation of natural resources, political instability for the *host* countries are some of the threats of globalization.

5.1.1 International Human Rights

To know what are the moral responsibilities and obligations of the multinational corporations in the host countries, let us discuss with the framework of rights ethics. Common minimum values should be followed to smoothen the transactions when the engineers and employers of MNCs have to operate at official, social, economic and sometimes political levels. At international level, the organizations are expected to adopt the minimum levels of (a) values, such as mutual support, loyalty, and (b) the negative duty of refraining from harmful actions such as violence and fraud, and ensuring fairness and practical justice in case of conflicts.

Intellectual Property Rights

Intellectual property right is a type of property right which allows the creators or owners of patents trademarks or copyrighted works to benefit from their own work or investment. These rights enable the right person to benefit from the protection of moral and material interests resulting from the authorship of scientific, literary or artistic productions. These rights are outlined in the article 27 of the Universal declaration of Human rights.

Protection of IPR

Like the other rights, the intellectual rights also should be protected and supported. The IPR (Intellectual property Rights) need to be protected in order to serve the following reasons

- The creations and inventions are the paths which lead to the progress of human development, either in technology or culture.
- These inventions should be protected legally in order to develop the commitment and interest for more creations.
- These intellectual properties must be protected and promoted which indirectly promote the economic growth that creates new jobs and industries, and enhances the quality and enjoyment of life.

The Intellectual property rights are protected by certain measures like patents, trademarks, industrial designs, copyrights, etc.

Industrial Designs

The ownership of an industrial design protects it from any duplication. Industrial designs are what make an article attractive and appealing and add commercial value to the product. This further increases marketability. Duplication will definitely mislead consumers and might also lead to the defamation of the original product.

Geographical Locations

The geographical location indications are helpful for the customers to identify the original and quality products, which are produced using the raw materials of that geographical area. This indication guarantees the customers that a product was produced in certain place and has certain characteristics that are due to the place of production.

5.7 ENGINEERS AS EXPERT WITNESS

Frequently engineers are required to act as consultants and provide expert opinion and views in many legal cases of the past events. They are required to explain the causes of accidents, malfunctions and other technological behavior of structures, machines, and instruments, e.g., personal injury while using an instrument, defective product, traffic accident, structure or building collapse, and damage to the property, are some of the cases where testimonies are needed. The focus is on the past.

The functions of eye-witness and expert-witness are different as presented in the Table 5.1.

Table 5.1 Eye-witness and expert-witness

<i>Eye-witness</i>	<i>Expert-witness</i>
1. Eye-witness gives evidence on only what has been seen or heard actually (perceived facts)	1. Gives expert view on the facts in their area of their expertise 2. Interprets the facts, in term of the cause and effect relationship 3. Comments on the view of the opposite side 4. Reports on the professional standards, especially on the precautions when the product is made or the service is provided

The engineers, who act as expert-witnesses, are likely to abuse their positions in the following manners:

1. Hired Guns

Mostly lawyers hire engineers to serve the interest of their clients. Lawyers are permitted and required to project the case in a way favorable to their clients. But the engineers have obligations to thoroughly examine the events and demonstrate their professional integrity to testify only the truth in the court. They do not serve the clients of the lawyers directly. The hired guns forward white lies and distortions, as demanded by the lawyers. They even withhold the information or shade the fact, to favor their clients.

2. Money Bias

Consultants may be influenced or prejudiced for monetary considerations, gain reputation and make a fortune.

3. Ego Bias

The assumption that the own side is innocent and the other side is guilty, is responsible for this behavior. An inordinate desire to serve one's client and get name and fame is another reason for this bias.

4. Sympathy Bias

Sympathy for the victim on the opposite side may upset the testimony. The integrity of the consultants will keep these biases away from the justice. The court also must obtain the balanced view of both sides, by examining the expert witnesses of lawyers on both sides, to remove a probable bias.

2.15. RELIGION

Religions have played major roles in shaping moral views and moral values, over geographical regions. Christianity has influenced the Western countries, Islam in the Middle-East countries, Buddhism and Hinduism in Asia, and Confucianism in China. Further, there is a strong psychological link between the moral and religious beliefs of people following various religions and faiths. Religions support moral responsibility. They have set high moral standards. Faith in the religions provides trust and this trust inspires people to be moral. The religions insist on tolerance and moral concern for others. Many professionals who possess religious beliefs are motivated to be morally responsible.

Each religion lays stress on certain high moral standards. For example, Hinduism holds polytheistic (many gods) view, and virtues of devotion and surrender to high order. Christianity believes in one deity and emphasizes on virtues of Love, Faith, and Hope. Buddhism is non-theistic and focuses on compassion and Islam on one deity and adherence of *ishan* (piety or pursuit of excellence) and prayer. Judaism stresses the virtue of 'tsedakah' (righteousness). But many religious sects have adopted poor moral standards, e.g., many religious sects do not recognize equal rights for women. The right to worship is denied for some people. People are killed in the name of or to promote religion. Thus, conflicts exist between the 'secular' and religious people and between one religion and another. Hence, religious views have to be morally scrutinized.

2.15.1 Divine Command Ethics

As per this principle, the right action is defined by the commands by God. It implies that to be moral, a person should believe in God and an action is right only if it is commanded by God. There are some difficulties in this approach, namely, (a) whether God exists or not is not clear. (b) How to know what are the God's commands? and (c) How to verify the genuineness of the commands? Further, religions such as Hinduism, Islam, and Christianity accept the existence of God. But Buddhism, Taoism, and Confucianism adopt only faith in a right path and do not believe in God.

Socrates was said to have argued that God, an entity which is responsible, morally good, and beyond fear or favor, would not command murder, rape, torture, immoral activities, and even mass suicide. Many such crimes were committed in the name of God then and continue even now in different parts of the world. Some Western leaders had claimed that God had commanded them to invade against the Middle-East countries. If anyone claims to have obtained commands from God to kill people merciless, then we have to conclude that the person is not religious but insane.

2.16 SELF-RESPECT

It is defined as valuing oneself in morally suitable ways. Self-respect includes (a) recognition, which means respect to others, their ideas, decisions, ability, and rights and (b) appraisal, which means properly valuing ourselves as to how well we face moral standards and our personal commitments (aims). An intensive but balanced feeling of self-respect is sense of honor. This includes intense agony and guilt for wrong doings. Self-control is a virtue of maintaining personal discipline (self-regulation). Courage is a by-product of self-respect, which makes a person face the hardship in rational way

Self-respect is different from self-esteem in the following manner:

<i>Self-respect</i>	<i>Self-esteem</i>
1. A moral concept	1. A psychological concept
2. Valuing oneself in morally-suitable ways	2. Having a positive attitude towards oneself. It may be excessive or unwarranted or normal
3. It includes virtues of recognition and appraisal. It promotes virtues of sense of honor, self-control and courage	

3.3 INDUSTRIAL STANDARDS

Industrial standards are important for any industry. Specification helps in achieving interchangeability. Standardization reduces the production costs and at the same time, the quality is achieved easily. It helps the manufacturer, customers and the public, in keeping competitiveness and ensuring quality simultaneously. Industrial standards are established by the Bureau of Indian Standards, in our country in consultation with leading industries and services.

International standards have become relevant with the development of the world trade. The International Standards Organization has now detailed specifications for generic products/services with procedures that the manufacturers or service providers should follow to assure the quality of their products or service. ISO 9000-2000 series are typical examples in this direction.

Table 3.1 gives a list of some types of standards with a few examples.

Table. 3.1 Industrial standards

<i>Aspects</i>	<i>Purpose</i>	<i>Examples</i>
1. Quality	Value appropriate to price	Surface finish of a plate, life of a motor
2. Quality of service	Assurance of product to ISO procedures	Quality of degrees according institutions by educational institutions
3. Safety	To safeguard against injury or damage to property	Methods of waste disposal
4. Uniformity of physical properties and functions	Interchangeability, ease of assembly	Standard bolts and nuts, standard time

Moonlighting

It is a situation when a person is working as employee for two different companies in the spare time. This is against the right to pursue one's legitimate self-interest. It will lead to conflict of interests, if the person works for competitors, suppliers or customers, while working under an employer. Another effect of moonlighting is that it leaves the person exhausted and harms the job performance in both places.

Insider Information

Another potential conflict of interest is when using 'inside' information to establish a business venture or get an advantage for oneself or one's family or friends. The information may be either of the parent company or its clients or its business partners, e.g., engineers might inform the decision on the company's merger with another company or acquisition or an innovative strategy adopted. In such cases, their friends get information on stock holding and decide on trading their stocks to sell or buy quickly, so that gain more or prevent a loss. For example, in WorldCom USA, the insider information was used to manipulate and sell a large amount of stock holding by the Director, upon knowing that the government has declined to admit their product.

4.9 COLLECTIVE BARGAINING

It is the bargain by the trade union for improving the economic interests of the worker members. The process includes negotiation, threatening verbally, and declaration of 'strike'. It is impossible to endorse fully the collective bargaining of unions or to condemn. There exist always conflicting views between the professionalism and unionism.

A. Faithful Agent or Trustee?

Professional societies such as NSPE and IETI refuse to accept the 'collective coercive action' of unionism, holding the principles of professional integrity as right, e.g., as per NSPE code III, i.e., engineers shall not promote their own interest at the expense of the dignity and integrity of the profession. The

engineers are said to exhibit a higher standard than self-interest; and they are expected to perform an ethical duty to their employer as faithful agent or trustee. The actions of unions are usually against the interests of the employers and they use coercion and force against the employers. These actions are interpreted as unprofessional and disloyal. But in certain cases, the safety of the workers had been ignored for a long period or the employees were under-paid for years. Can we still hold the action as unethical?

It can be concluded from this discussion, that

- (a) The duty of the employee to one's employer does not mean sacrifice of monetary self-interests, and
- (b) trustee or faithful agent means executing the assigned tasks and safeguarding the property. It does not nullify the right to negotiate for safe and hygienic working conditions, and economic benefits collectively.
- (c) The codes insist that the paramount obligation is to the society, as compared to their employers. The duty to the employers is also limited by considerations such as workers safety, and the right to disobey illegal or unethical activities. After all the employees are also parts of the society, and
- (d) Can collective and coercive action be resorted, when all other efforts have failed?

B. Service to the Public?

The service to the public is of foremost importance. But the unions promote the interests of a few members only. The public welfare should not suffer because of their actions. Imagine a situation when all the teachers, medical practitioners, and ambulance drivers go on strike. Will this not cause damage to the public safety and health?

Collective bargaining by engineers through union or association or forums may act within limits set by the concern for the public welfare. Professional societies can play a great role in the promotion and establishment of principles and practices towards fair employment and exploitation. But they can not function as collective bargaining agents.

The collective bargaining can not be judged as unethical, unless we study the cases individually and decide. The collective bargaining is acceptable per se, but the means should be constructive, persuasive, firm based on mutual understanding, and not destructive, disruptive, and not harming the persons or property.

C. Assessment on Unionism

The moral assessment on Unions is a complex process. A careful consideration of all relevant moral facts are to be inquired into and judged. It can not be generalized, because of the divergent views on unionism, as shown in Table. 4.1.

Table. 4.1 Pro- and anti-views on unionism

For unionism	Against unionism
1. Unions have been useful in improving the standard of living and economic benefits of the workers. Even non-union members leading to inflationary condition are able to get those benefits.	1. Unions have lead to disturb the economy of state by salaries, and increase salaries and expenses, leading to inflationary conditions.
2. Unions have obtained greater participation in organization, by participative management. Union members are appointed as Directors in the Board and credited to act as bridge between the employers and employees.	2. Instead of being cooperative, they act in negative and destructive ways, causing loss of man-days. Opinions of the individual worker is suppressed and used as pawns.
3. Unions have contributed to the job security, and protection against arbitrary treatment to the employees.	3. Unions encourage mediocrity, and act in favor of seniority-based promotion. Merit-based promotion and awards for personal achievement are disregarded
4. They are able to put resistance to unethical orders and support to ethical actions	4. Unions thrive on prolonged unrest, dissatisfied, and tense relations between workers and management.
5. They have provided for effective grievance redressal mechanism for employees.	5. They cause pigeon-holing of employee in narrow job classifications to which the salary scales are attached.
6. They act to safeguard against the possible political interference, exploitation, and alienation in the company affairs.	

5.5 ENGINEERS AS MANAGERS

5.5.1 Characteristics

The characteristics of engineers as managers are:

1. Promote an ethical climate, through framing organization policies, responsibilities and by personal attitudes and obligations.
2. Resolving conflicts, by evolving priority, developing mutual understanding, generating various alternative solutions to problems.
3. Social responsibility to stakeholders, customers and employers. They act to develop wealth as well as the welfare of the society. Ethicists project the view that the manager's responsibility is only to increase the profit of the organization, and only the engineers have the responsibility to protect the safety, health, and welfare of the public. But managers have the ethical responsibility to produce safe and good products (or useful service), while showing respect for the human beings who include the employees, customers and the public. Hence, the objective for the managers and engineers is to produce valuable products that are also profitable.

5.5.2 Managing Conflicts

In solving conflicts, force should not be resorted. In fact, the conflict situations should be tolerated, understood, and resolved by participation by all the concerned. The conflicts in case of project managers arise in the following manners:

- (a) Conflicts based on schedules: This happens because of various levels of execution, priority and limitations of each level.
- (b) Conflicts arising out of fixing the priority to different projects or departments. This is to be arrived at from the end requirements and it may change from time to time.
- (c) Conflict based on the availability of personnel.
- (d) Conflict over technical, economic, and time factors such as cost, time, and performance level.
- (e) Conflict arising in administration such as authority, responsibility, accountability, and logistics required.
- (f) Conflicts of personality, human psychology and ego problems.
- (g) Conflict over expenditure and its deviations.

Most of the conflicts can be resolved by following the principles listed here:

1. *People*

Separate people from the problem. It implies that the views of all concerned should be obtained. The questions such as what, why, and when the error was committed is more important than to know who committed it. This impersonal approach will lead to not only early solution but also others will be prevented from committing errors.

2. *Interests*

Focus must be only on interest i.e., the ethical attitudes or motives and not on the positions (i.e., stated views). A supplier may require commission larger than usual prevailing rate for an agricultural product.

But the past analysis may tell us that the material is not cultivated regularly and the monsoon poses some additional risk towards the supply. Mutual interests must be respected to a maximum level. What is right is more important than who is right!

3. *Options*

Generate various options as solutions to the problem. This helps a manager to try the next best solution should the first one fails. Decision on alternate solutions can be taken more easily and without loss of time.

4. *Evaluation*

The evaluation of the results should be based on some specified objectives such as efficiency, quality, and customer satisfaction. More important is that the means, not only the goals, should be ethical.

4.16 INTELLECTUAL PROPERTY RIGHTS

Patent, Publish, and Prosper — Dr. Mashelkar, Dir. Genl., CSIR

Future can be best predicted by inventing it — Xerox, USA

4.16.1 Intellectual Property

It is the information and original expression that derives its original value from creative ideas, and is with a commercial value. IP permits people to have fully independent ownership for their innovations

and creativity, like that of own physical property. This encourages the IP owners towards innovation and benefit to the society. It is an asset that can be bought or sold, licensed, and exchanged. It is intangible i.e., it cannot be identified by specific parameters.¹¹

The agreements with World Trade Organisation (WTO) and Trade-Related aspects of Intellectual Property System (TRIPS) have been adopted effective from January 2005. Besides the minimum standards set for protection of IP rights, appropriate laws framed by the member countries are expected to reduce distortions and barriers for and promote the international trade. The global IPR system strengthens protection, increases the incentives for innovation, and raises returns on international technology transfer. However, it could raise the costs of acquiring new technology and products, shifting the global terms of trade in favor of technology producers.

The agreements establish norms and conditions for the following instruments of intellectual properties:

1. Patents

Patent is a contract between the individual (inventor) and the society (all others). Patents protect legally the specific products from being manufactured or sold by others, without permission of the patent holder. Patent holder has the legally-protected monopoly power as one's own property. The validity is 20 years from the date filing the application for the patent. It is a territorial right and needs registration. The Patent (Amendment) Act 2002 guarantees such provisions.

Patent is given to a product or a process, provided it is entirely new, involving an inventive method and suitable for industrial application. While applying for a patent, it is essential to submit the documents in detail regarding the problem addressed, its solution, extent of novelty or innovation, typical applications, particulars of the inventor, and the resources utilized. Inventions are patentable and the discoveries are not.

Some of the salient features of the Patent Act 2002 are listed as follows:

1. **Patent outside India:** Applications for Patent outside India, on inventions for defense purposes or related to atomic energy are prohibited. For other patents, an inventor should apply for a patent within India and then seek clearance for filing patents abroad.
2. **Licensing:** The controller of patents grants the license upon verification and on some terms and conditions. The controller shall endeavor to secure that the articles manufactured under the patent shall be available to the public at the lowest price consistent with patentees deriving a reasonable gain from their patent rights.
3. **Negative right:** The grant of patent for an invention does not guarantee the merit or any other commercial value of the invention disclosed. The state which grants the patent does not also guarantee the validity of the patent granted. If other regulations do not permit, even the patent holder can not commence manufacturing. In this context, it is a negative right.
4. **First to file rule:** Indian like many other countries follows the system of first to file or first to register system to determine priority. Accordingly, a patent or invention which is filed or registered first in the patent office will have precedence to the patent or invention, which is filed later in the date, even if it had been invented earlier.
5. **Burden of proof of infringement:** Legal rights of patent can not be enforced automatically. In any suit for infringement of a patent, the patentee must move the court. The court may direct the defendant to prove that the process used by him to obtain the product, identical to the product of the patented process, is different from the patented process.

Types of Patents

(a) Utility Patent

The utility patent is granted to anyone who invents or discovers any new and useful process, machine, manufacture or chemical composition of any manner or any new and useful improvement thereof. The utility time is 20 years.

(b) Industrial Design Patent

The industrial design patent is an idea or conception regarding features of shape, configuration, pattern, ornamental with lines or colors applied to any article, two or three dimensional, made by any industrial process and is judged by the eye or a product. The Designs Act 2000 excludes from its purview the functioning features of an article and grants protection only to those which have an aesthetic appeal. For example, the design of a tea cup must have a hollow receptacle for holding tea and a handle to hold the cup. These are functional features that can not be registered. But a fancy shape or ornamentation on it would be registerable. A table has a flat surface on which other objects can be placed. This is its functional element. But its shape, color or the way it is supported by legs or otherwise, are all elements of design or artistic elements and they are registerable, if unique and novel. Other examples include the design applied to shoes, T.V., and textiles.

The design patent has a term of 14 years from the date of filing the application. Designs Act 2000 gives further details on this aspect.

2. Copyright

The copyright is a specific and exclusive right, describing rights given to creators for their literary and artistic works. This protects literary material, aesthetic material, music, film, sound recording, broadcasting, software, multimedia, paintings, sculptures, and drawings including maps, diagrams, engravings or photographs. There is no need for registration and no need to seek lawyer's help for settlement. The life of the copyright protection is the life of the inventor or author plus 50 years.

Copyright gives protection to particular expression and not for the idea. Copyright is effective in (a) preventing others from copying or reproducing or storing the work, (b) publishing and selling the copies, (c) performing the work in public, commercially (d) to make film (e) to make translation of the work, and (f) to make any adaptation of the work. Copying the idea is called 'plagiarism' and it is dealt with separately.

Can software be protected through copyright? Indian copyright Act amended in 1984 included the rights of in a computer program as literary work. Many countries protect software as a copyright. Some holds the view that copyright is not the right type of protection for software. They held that the patents and trade secrets are more appropriate forms of protecting software. While trade secret is the most conventional form of protection of software, in the recent years, both patents and copyrights are adopted to protect software.¹¹

Copyright (Amendment) Act 1999, India ensures fair dealing of broadcasting through the internet. The concerns of Book industry, Music Industry, Film and Television Industry, Computer Industry and Database Industry are sufficiently met by this updated Act.

3. Trademark

Trademark is a wide identity of specific good and services, permitting differences to be made among different trades. It is a territorial right, which needs registration. Registration is valid initially for 10 years, and renewable. The trademark or service mark may be registered in the form of a device, a heading, a label, a ticket, a letter, a word or words, a numeral or any combination of these, logos, designs, sounds, and symbols. Trademark should not be mistaken for a design, e.g., the shape of a bottle in which a product is marketed, can not be registered as a trademark. Trademarks Act 1999 made in compliance with TRIPS agreement, provides further details.¹¹

There are three functions of trademark:

1. Just as we are identified by our names, good are identified by their trademarks. For example, the customer goes to the shop and asks for Lux soap. The word 'Lux' is a trade mark. In other words it shows the origin or source of the goods.
2. The trademark carries with it an inherent indication or impression on the quality of goods, which indirectly demonstrates that it receives the customer's satisfaction.
3. The trademark serves as silent sales promoter. Without a trademark, there can be no advertisement. In other words, it serves as a medium for advertising the goods.

The marks should be distinctive i.e., it should be able to distinguish from one good to the other. The terms used for trademarks are usually generic, descriptive, and suggestive. Some of the term which are not distinctly distinguishing the goods or services from others, are called *generic term* and are eligible for protection under trademarks. The descriptive term should clearly indicate or convey the specific purpose, function, physical characteristic and the end use of the product. Relatively, the suggestive

marks do not describe the goods at first sight, but with an element of imagination or perception the nature of the goods can be understood. Thus, the suggestive marks are distinctive and are protected as trademarks. Arbitrary marks and fanciful marks are distinctive and hence accepted for registration.

Besides this, there is also a certification mark by the Bureau of Indian Standards (BIS or ISI) which guarantees that the holder's product bearing the mark has met certain standards or requirements. This adds considerably to the market value and to a great value in the export trade.

4. Trade Secret

A trade secret is the information which is kept confidential as a secret. This information is not accessed by the any other (competitor) than the owner and this gives a commercial advantage over the competitors. The trade secrets are not registered but only kept confidential. These are given limited legal protection, against abuse by the employee or contractor, by keeping confidentiality and trust.

The trade secrets may be formulae, or methods, or programs, or processes or test results or data collected, analyzed, and synthesized. These are related to designs, technical processes, plant facilities, list of suppliers or customers etc. This information should not be disclosed or used by any other person.

2.11 THEORIES ABOUT RIGHT ACTION (ETHICAL THEORIES)

2.11.1 Uses and Criteria

The ethical theories are useful in many respects.

1. In understanding moral dilemma. They provide clarity, consistency, systematic and comprehensive understanding.
2. It provides helpful practical guidance in moral issues towards the solution.
3. Justifying professional obligations and decisions, and
3. In relating ordinary and professional morality.

Different *criteria* may be applied for evaluating various ethical theories and deciding upon the best.

1. The theory must be clear and (coherent) formulated with concepts that are logically connected.
2. It must be internally consistent, i.e., none of its principles conflicts with any other
3. The theory and its defense must depend, only upon facts.
4. It must organize basic moral values in systematic and comprehensive manner. It is to fix priority of values and provide guidance in all situations
5. It must provide guidance compatible with our moral convictions (judgments) about concrete situations. For example, if an ethical theory says that it is all right for engineers to make explosive devices without the informed consent of the public, we can conclude that the theory is inadequate.

Theories and judgments are continually adjusted to each other until we reach a reflective equilibrium. Most of the theories converge towards the welfare of the humanity. The duty ethics and right ethics differ in great extent on their emphasis. But they remain complementary always.

2.11.2 Ethical Theories/Approaches

Several ethical theories have been developed over different times, each of them stressing certain ethical principles or features. Each stresses a view and many a times, we find that these theories converge and reinforce the ethics, in deciding upon the actions and justifying the results.

- ✖ **Duty Ethics**
- ✖ **Rights Ethics**
- ✖ **Virtue Ethics**

50

UTILITARIANISM JOHN STUART MILL (1806-1873)

- ✖ **Considers a balance of good & bad consequences for everyone affected (society)**
- ✖ **Actions are good that serve to promote human well-being**
- ✖ **Cost-Benefit analysis is an application**
- ✖ **Consideration of most benefit to the most people outweighs needs of a few individuals**

51

DUTY ETHICS IMMANUEL KANT (1724-1804)

- ✖ **There are duties that should be performed (e.g.. Duty to treat others fairly or not to injure others) regardless of whether these acts do the most good or not.**

52

RIGHTS ETHICS JOHN LOCKE (1632-1704)

- ✖ **People have fundamental rights (like life, liberty, & property) that others have a duty to respect.**

53

TESTING ETHICAL THEORIES

- ✖ **Five widely used tests for evaluating ethical theories:**
 - + **The theory must be clear**
 - + **It must be consistent**
 - + **Neither the theory nor its defense can rely upon false information**
 - + **It must be sufficiently comprehensive to provide guidance in specific situations of interest to us.**
 - + **It must be compatible with our most carefully considered moral convictions about concrete situations.**

54

SELF-INTEREST AND ETHICAL EGOISM

- ✖ **is being good and acceptable to**

51-54/61



1. Utilitarian Theory

The term Utilitarianism was conceived in the 19th century by **Jeremy Bentham** and **John Stuart Mill** to help legislators determine which laws were morally best. They suggested that the standard of right conduct is maximization of good consequences. Good consequences mean either 'utilities' or the 'balance of good over evil'. This approach weighs the costs and benefits. Right actions are the ones that produce the greatest satisfaction of the preferences of the affected persons. In analyzing an issue in this approach, we have to:

- (a) Identify the various courses of action available to us.
- (b) Ask who will be affected by each action and what benefits or harms will be derived from each.
- (c) Choose the action that will produce the greatest benefits and the least harm. The ethical action is the one that provides the greatest good for the greatest number.

The ACT UTILITARIAN theory proposed by **J.S. Mill** (1806-73) focuses on actions, rather than on general rules. An action is right, if it generates the most overall good for the most people involved.

The RULE UTILITARIAN theory, developed by **Richard Brandt** (1910-97), stressed on the rules, such as 'do not steal', 'do no harm others', 'do not bribe', as of primary importance. He suggested that individual actions are right when they are required by set of rules which maximizes the public good.

The act utilitarian theory permitted a few immoral actions. Hence, there was need to develop rule *utilitarian theory* to establish morality and justice, in the transactions. For example, stealing an old computer from the employer will benefit the employee more than the loss to the employer. As per Act, utilitarian this action is right. But rule utilitarian observes this as wrong, because the employee should act as 'faithful agent or trustee of the employees'. In another example, some undisciplined engineers are terminated with the blame for the mistakes they have not committed.

The process is unfair although this results in promotion of overall good.

selling oneself to slavery.

The Welfare Rights are the rights to benefit the needy for a decent human life, when one can not earn those benefits and when those benefits are available in the society.

- E. *Economic rights*: In the free-market economy, the very purpose of the existence of the manufacturer, the sellers and the service providers is to serve the consumer. The consumer is eligible to exercise some rights⁹. The consumers' six basic rights are: Right to Information, Right to Safety, Right to Choice, Right to be Heard, Right to Redressal, and Right to Consumer Education.

A few rights are absolute, i.e., unlimited and have no justifiable exceptions. For example, rights ethicists view that the rights have not been violated if the people purchase a (technological product) hang glider and they get injured by flying them carelessly or under bad weather conditions. But human rights imply that one not to be poisoned or killed by technological products, whose dangers are not obvious or wantonly hidden. They imply a right to be informed, when the purchase was made, of the possible dangers during use or service (obtaining informed consent).

Rights ethics is distinctive in that it makes human rights the ultimate appeal — the moral bottom line. Human rights constitute a moral authority to make legitimate moral demands on others to respect

our choices, recognizing that others can make similar claims on us. Thus, we see that the rights ethics provides a powerful foundation for the special ethical requirements in engineering and other professions.

4. The Virtue Theory

This emphasizes on the character rather than the rights or duties. The character is the pattern of virtues (morally-desirable features). The theory advocated by Aristotle, stressed on the tendency to act at proper balance between extremes of conduct, emotion, desire, attitudes to find the golden mean between the extremes of 'excess' or 'deficiency'. The examples shown below illustrate the theory:

<i>Virtue</i>	<i>Excess</i>	<i>Golden mean</i>	<i>Deficient</i>
Truthfulness (governs communication)	Revealing all in violation of tact and confidentiality	Necessary and sufficient, to proper person	Secretive
Courage (face danger, risk)	Roguishness, bold	Firm and humble	Cowardice
Generosity (giving)	Wasting resources	Give, in appropriate measure	Miserly
Friendliness (governs relationship)	Without anger, effusive	Within decent limits	Bad-tempered
Green environment	Exploitation	Protection	Neglect
Work and earn	Tiresome work (strained)	Balance of work and leisure	Lazy (no work) and more pay

On the other hand, the Virtue Theory proposed by **Mac Intyre**, highlighted on the actions aimed at achieving common good and social (internal) good such as social justice, promotion of health, creation of useful and safe technological products and services. Five types of virtues that constitute responsible professionalism, namely public-spirited virtues, proficiency virtues, team-work virtues, self-governance virtues, and cardinal virtues are discussed in # 2.10.3.

5. Self-realisation Ethics

Right action consists in seeking self-fulfillment. In one version of this theory, the self to be realized is defined by caring relationships with other individuals and society. In another version called *ethical egoism*, the right action consists in always promoting what is good for oneself. No caring and society relationships are assumed.

6. Justice (Fairness) Theory

The justice or fairness approach to ethics has its roots in the teachings of the ancient Greek philosopher Aristotle, who said that "equals should be treated equally and unequals unequally." The basic moral question in this approach is: How fair is an action? Does it treat everyone in the same way, or does it show favoritism and discrimination?

Environmental ethics is the study of (a) moral issues concerning the environment, and (b) moral perspectives, beliefs, or attitudes concerning those issues.

Engineers in the past are known for their negligence of environment, in their activities. It has become important now that engineers design eco-friendly tools, machines, sustainable products, processes, and projects. These are essential now to (a) ensure protection (safety) of environment (b) prevent the degradation of environment, and (c) slow down the exploitation of the natural resources, so that the future generation can survive.

The American Society of Civil Engineers (ASCE) code of ethics, has specifically requires that “engineers shall hold paramount the safety, health, and welfare of the public and shall strive to comply with the principles of sustainable development in the performance of professional duties” The term *sustainable development* emphasizes on the investment, orientation of technology, development and functioning of organizations to meet the present needs of people and at the same time ensuring the future generations to meet their needs.

Compaq Computer Corporation (now merged with HP) was the leader, who exhibited their commitment to environmental health, through implementation of the concept of ‘Design for environment’ on their products, unified standards all over the world units, and giving priority to vendors with a record of environmental concern.

Engineers as experimenters have certain duties towards environmental ethics, namely:

1. *Environmental impact assessment*: One major but sure and unintended effect of technology is wastage and the resulting pollution of land, water, air and even space. Study how the industry and technology affects the environment.
2. *Establish standards*: Study and to fix the tolerable and actual pollution levels.
3. *Counter measures*: Study what the protective or eliminating measures are available for immediate implementation
4. *Environmental awareness*: Study on how to educate the people on environmental practices, issues, and possible remedies.

5.2.1 Disasters

1. Plastic Waste Disposal

In our country, several crores of plastic bottles are used as containers for water and oil, and plastic bags are used to pack different materials ranging from vegetables to gold ornaments. Hardly any of these are recycled. They end up in gutters, roadsides, and agricultural fields. In all these destinations, they created havoc. The worse still is the burning of plastic materials in streets and camphor along with plastic cover in temples, since they release toxic fumes and threaten seriously the air quality. Cities and local administration have to act on this, collect and arrange for recycling through industries.

2. e-Waste Disposal

The parts of computers and electronic devices which have served its useful life present a major environmental issue for all the developing countries including India. This scrap contains highly toxic elements such as lead, cadmium, and mercury.

Even the radioactive waste will lose 89% of its toxicity after 200 years, by which time it will be no more toxic than some natural minerals in the ground. It will lose 99% of its remaining toxicity over the next 30,000 years. The toxic chemical agents such as mercury, arsenic, and cadmium retain toxicity undiminished for ever.

But these scraps are illegally imported by unscrupulous agencies to salvage some commercially-valuable inputs. Instead of spending and managing on the scrap, unethical organizations sell them to countries such as India. This is strictly in violation of the Basel Convention of the United Nations Environment Program, which has banned the movement of hazardous waste. A recent report of the British Environment Agency,¹³ has revealed that the discarded computers, television sets, refrigerators, mobile phones, and electrical equipments have been dispatched to India and Pakistan in large quantity, for ultimate disposal in environmentally-unacceptable ways and at great risk to the health of the labour. Even in the West, the electronic junk has been posing problems. Strong regulation including (a) pressure on industries to set up disassembling facilities, (b) ban on disposal in landfill sites, (c) legislation for recycling requirements for these junk and (d) policy incentives for eco-friendly design are essential for our country. The European Union through the Waste Electrical and Electronic Equipment (WEEE)

directive has curbed the e-waste dumping by member countries and require manufacturers to implement methods to recover and recycle the components.

Indian Government expressed its concern through a technical guide on environmental management for IT Industry in December, 2004. It is yet to ratify the ban on movement of hazardous waste according to the Basel Convention. A foreign news agency exposed a few years back, the existence of a thriving e-waste disposal hub in a suburb of New Delhi, operating in appallingly dangerous conditions. Our country needs regulations to define waste, measures to stop illegal imports, and institutional structures to handle safe disposal of domestic industrial scrap.

3. Industrial Waste Disposal

There has been a lot of complaints through the media, on (a) against the Sterlite Copper Smelting Plant in Thuthukkudi (1997) against its pollution, and (b) when Indian companies imported the discarded French Warship *Clemenceau* for disposal, the poisonous *asbestos* compounds were expected to pollute the atmosphere besides exposing the labor to a great risk, during the disposal. The government did not act immediately. Fortunately for Indians, the French Government intervened and withdrew the ship, and the serious threat was averted!

4. Depletion of Ozone Layer

The *ozone* layer protects the entire planet from the ill-effects of ultraviolet radiation and is vital for all living organisms in this world. But it is eaten away by the Chloro-fluro-carbons (CFC) such as *freon* emanating from the refrigerators, air conditioners, and aerosol can spray. This has caused also skin cancer to sun-bathers in the Western countries. Further NO and NO₂ gases were also found to react with the ozone. Apart from engineers, the organizations, laws of the country and local administration and market mechanisms are required to take up concerted efforts to protect the environment.

5. Global Warming

Over the past 30 years, the Earth has warmed by 0.6 °C. Over the last 100 years, it has warmed by 0.8 °C. It is likely to push up temperature by 3 °C by 2100, according to NASA's studies. The U.S. administration has accepted the reality of global climate change, which has been associated with stronger hurricanes, severe droughts, intense heat waves and the melting of polar ice. Greenhouse gases, notably carbon dioxide emitted by motor vehicles and coal-fired power plants, trap heat like the glass walls of a greenhouse, cause the Earth to warm up. Delegates from the six countries — Australia, China, India, Japan, South Korea and US met in California in April 2006 for the first working session of the Asia-Pacific Partnership on Clean Development and Climate. These six countries account for about half of the world's emissions of climate-heating greenhouse gases. Only one of the six, Japan, is committed to reducing greenhouse gas emissions by at least 5.2 per cent below 1990 levels by 2012 under the Kyoto Agreement.

About 190 nations met in Germany in the middle of May 2006 and tried to bridge vast policy gaps between the United States and its main allies over how to combat climate change amid growing evidence that the world is warming that could wreak havoc by stoking more droughts, heat waves, floods, more powerful storms and raise global sea levels by almost a meter by 2100.

6. Acid Rain

Large emissions of sulphur oxides and nitrous oxides are being released in to the air from the thermal power stations using the fossil fuels, and several processing industries. These gases form compounds with water in the air and precipitates as rain or snow on to the earth. The acid rain in some parts of the world has caused sufficient damage to the fertility of the land and to the human beings.

5.2.2 Human-centered Environmental Ethics

This approach assumes that only human beings have inherent moral worth duly to be taken care of. Other living being and ecosystems are only instrumental in nature. Utilitarianism aims to maximize good consequences for human beings. Most of the goods are engineered products made out of natural resources. Human beings have also (a) recreational interests (enjoy leisure through mountaineering, sports, and pastimes), (b) aesthetic interests (enjoy nature as from seeing waterfalls and snow-clad mountains), (c) scientific interests to explore into nature or processes, and (d) a basic interest to survive, by preservation as well as conservation of nature and natural resources.

Rights ethicists favor the basic rights to live and right to liberty, to realise the right to a live in a supportive environment. Further, virtue ethics stresses importance of prudence, humility, appreciation of natural beauty, and gratitude to the mother nature that provides everything.

However, the nature-centered ethics, which ensures the worth of all living beings and organisms, seems to be more appropriate in the present-day context. Many Asian religions stress the unity with nature, rather than domination and exploitation. The Zen Buddhism calls for a simple life with compassion towards humans and other animals. Hinduism enshrines the ideal of oneness (*advaita*) in and principle of *ahimsa* to all living beings. It identifies all the human beings, animals, and plants as divine. The eco-balance is the need of the hour and the engineers are the right experimenters to achieve this.

Computer ethics is defined as (a) study and analysis of nature and social impact of computer technology, (b) formulation and justification of policies, for ethical use of computers. This subject has become relevant to the professionals such as designers of computers, programmers, system analysts, system managers, and operators. The use of computers have raised a host of moral concerns such as free speech, privacy, intellectual property right, and physical as well as mental harm. There appears to be no conceptual framework available on ethics, to study and understand and resolve the problems in computer technology.

5.3.1 Types of Issues

Different types of problems are found in computer ethics.

1. Computer as the Instrument of Unethical Acts

- (a) The usage of computer replaces the job positions. This has been overcome to a large extent by readjusting work assignments, and training everyone on computer applications such as word processing, editing, and graphics.

- (b) Breaking privacy. Information or data of the individuals accessed or erased or the ownership changed.
- (c) Defraud a bank or a client, by accessing and withdrawing money from other's bank account.

2. Computer as the Object of Unethical Act

The data are accessed and deleted or changed.

- (a) *Hacking*: The software is stolen or information is accessed from other computers. This may cause financial loss to the business or violation of privacy rights of the individuals or business. In case of defense information being hacked, this may endanger the security of the nation.
- (b) *Spreading virus*: Through mail or otherwise, other computers are accessed and the files are erased or contents changed altogether. 'Trojan horses' are implanted to distort the messages and files beyond recovery. This again causes financial loss or mental torture to the individuals. Some hackers feel that they have justified their right of free information or they do it for fun. However, these acts are certainly unethical.
- (c) *Health hazard*: The computers pose threat during their use as well as during disposal. These are discussed in # 5.3.2 and # 5.2.1, respectively, in detail.

3. Problems Related to the Autonomous Nature of Computer

- (a) *Security risk*: Recently the Tokyo Stock Exchange faced a major embarrassment. A seemingly casual mistake by a junior trader of a large security house led to huge losses including that of reputation. The order through the exchange's trading system was to sell one share for 600,000 Yen. Instead the trader keyed in a sale order for 600,000 shares at the rate of one Yen each. Naturally the shares on offer at the ridiculously low price were lapped up. And only a few buyers agreed to reverse the deal! The loss to the securities firm was said to be huge, running into several hundred thousands. More important to note, such an obvious mistake could not be corrected by some of the advanced technology available. For advanced countries like Japan who have imbibed the latest technology, this would be a new kind of learning experience.¹²

- (b) *Loss of human lives*: Risk and loss of human lives lost by computer, in the operational control of military weapons. There is a dangerous instability in automated defense system. An unexpected error in the software or hardware or a conflict during interfacing between the two, may trigger a serious attack and cause irreparable human loss before the error is traced. The Chinese embassy was bombed by U.S. military in Iraq a few years back, but enquiries revealed that the building was shown in a previous map as the building where insurgents stayed.
- (c) In flexible manufacturing systems, the autonomous computer is beneficial in obtaining continuous monitoring and automatic control.

Various issues related to computer ethics are discussed as follows:

5.3.2 Computers In Workplace

The ethical problems initiated by computers in the workplace are:

1. Elimination of routine and manual jobs. This leads to unemployment, but the creation of skilled and IT-enabled service jobs are more advantageous for the people. Initially this may

require some upgradation of their skills and knowledge, but a formal training will set this problem right. For example, in place of a typist, we have a programmer or an accountant.

2. *Health and safety*: The ill-effects due to electromagnetic radiation, especially on women and pregnant employees, mental stress, wrist problem known as *Carpel Tunnel Syndrome*, and backpain due to poor ergonomic seating designs, and eye strain due to poor lighting and flickers in the display and long exposure, have been reported worldwide. Over a period of long exposure, these are expected to affect the health and safety of the people. The computer designers should take care of these aspects and management should monitor the health and safety of the computer personnel.
3. *Computer failure*: Failure in computers may be due to errors in the hardware or software. Hardware errors are rare and they can be solved easily and quickly. But software errors are very serious as they can stop the entire network. Testing and quality systems for software have gained relevance and importance in the recent past, to avoid or minimize these errors.